

FoodPulse AI

Smart Energy & Process Optimization for Food-Processing SMEs

Executive summary. FoodPulse AI is a retrofit-first energy and process intelligence layer for Indian food-processing SMEs. It combines energy/process sensing, production context, expected-energy modeling, anomaly detection, equipment-health scoring and constrained process optimization. The system does not simply display consumption; it connects energy deviations to operating conditions and converts them into actionable recommendations with estimated savings and emissions impact.

Prototype result. In a representative simulated food-processing plant, specific energy consumption is reduced from **2.50 to 2.18 kWh/kg**, a **12.8% reduction**, while production remains 5,000 kg/day and simulated quality remains 96%. These are simulation values for the prototype and must be replaced by measured site data during a pilot.

1. Problem and Indian SME fit

Food-processing SMEs often operate mixed-age equipment, have limited machine-level energy visibility and need efficiency interventions that do not disrupt production. BEE's current MSME work explicitly includes Food Processing among the sectors receiving energy-efficiency support, and its ADEETIE scheme is designed around energy-efficient technology deployment, energy audits, project support and monitoring/verification. BEE also describes low-cost technology and operating practices as important adoption considerations for MSMEs.

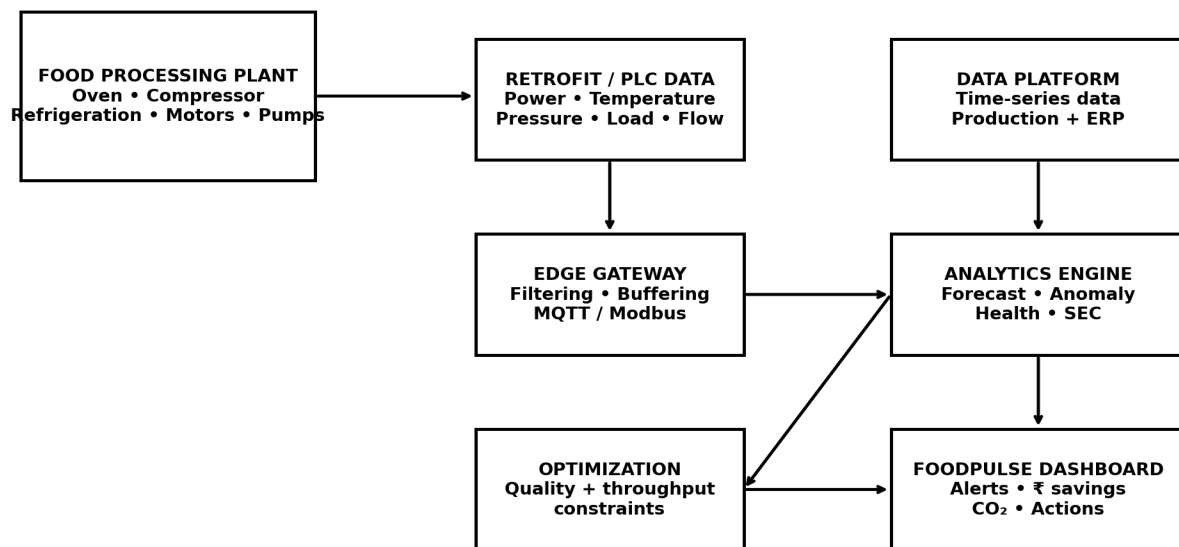
FoodPulse therefore uses a retrofit-first architecture: existing machines remain in service, while meters and process sensors provide the data layer. The first deployment can be limited to the highest-energy assets—oven/dryer, refrigeration, compressor, motors and pumps—rather than requiring a full Industry 4.0 transformation.

2. Solution mechanism

Step	Mechanism
Collect	Power, temperature, pressure, load, production and quality data from meters, PLCs or retrofit sensors
Normalize	Associate energy consumption with production output, operating hours and process conditions.
Baseline	Train an expected-energy model using normal historical operation.
Detect	Flag periods where actual energy exceeds expected energy beyond a configurable threshold.
Diagnose	Correlate the deviation with equipment signals and process conditions.
Optimize	Recommend operating points subject to production and quality constraints.
Verify	Compare post-intervention SEC against the baseline and quantify energy, cost and emissions savings

3. Technical architecture

FoodPulse AI — System Architecture



Plant equipment can expose data through existing PLC/SCADA interfaces or low-cost retrofit meters and sensors. An edge gateway performs filtering, local buffering and protocol translation (for example Modbus/MQTT). The analytics layer consumes time-series measurements plus production/ERP information. The recommendation layer returns alerts and operating actions to the dashboard; closed-loop machine control is optional and should only be enabled after plant validation.

4. Data and analytics

Primary KPI: SEC = Total energy consumed / production output, expressed as kWh/kg. The energy model predicts expected energy from production quantity, ambient conditions, oven temperature, compressor pressure, cold-room temperature, motor load and time-of-day. A deviation score is calculated as (actual - expected) / expected. Equipment health combines energy deviation with process-signal deviations.

The prototype includes simulated compressor degradation, refrigeration inefficiency, oven inefficiency and idle-energy waste. This makes the demonstration reproducible without physical hardware.

5. Quantified improvement

Metric	Baseline	Optimized	Change
Production	5,000 kg/day	5,000 kg/day	Preserved
Energy	12,500 kWh/day	10,900 kWh/day	−1,600 kWh/day
SEC	2.50 kWh/kg	2.18 kWh/kg	−12.8%
Quality	96%	96%	Preserved

The optimization is constrained by a minimum quality score of 95% and a fixed production target. This prevents the system from claiming energy savings by simply reducing output or degrading product quality.

6. Business and deployment model

Target segment: small and medium food processors such as bakeries/snacks, dairy, frozen foods, spice processing and similar batch/continuous operations with meaningful electricity or thermal-energy loads.

Indicative pilot cost: approximately ₹1 lakh for a small site, covering a starter set of energy meters/CTs, temperature/pressure sensors, edge gateway, installation and software setup. Indicative SaaS:

₹3,000–₹8,000/month/site. These figures are planning assumptions and should be validated with vendors and site scope.

Payback: calculate from measured baseline savings rather than promising a fixed period. Example: if a pilot site realizes ₹25,000/month of verified savings, a ₹1 lakh installation has a simple payback of about four months. Larger sites can justify additional sensing and process controls.

Scale-up: pilot one site → verify SEC reduction → standardize a retrofit kit → deploy across food-processing clusters → add ERP/MES integration → aggregate multi-site benchmarking.

7. Key assumptions and limitations

The prototype dataset is synthetic and is not presented as measured plant data. Electricity tariff, emissions factor, production scale and quality response curves are configurable assumptions. A real deployment requires an energy audit, meter validation, product-quality acceptance criteria, cybersecurity review, and a measurement & verification plan. Recommendations should initially be advisory rather than automatic control commands.

8. Why this fits the challenge

FoodPulse addresses the challenge through four connected outcomes: (1) lower specific energy consumption, (2) real-time visibility, (3) quantified decarbonization opportunities, and (4) a retrofit-first business model suitable for SMEs. The design also supports the measurement-and-verification approach used in energy-efficiency projects: savings are demonstrated through the difference between baseline and post-intervention SEC at comparable production conditions.

References

Bureau of Energy Efficiency (BEE), ADEETIE — Assistance in Deploying Energy Efficient Technologies in Industries & Establishments.

BEE, Energy Efficiency in SMEs / Small Medium Enterprise programme.

BEE, Energy & Resource Mapping Study — Food Processing and other MSME clusters.

BEE, Sectoral Report: Food Processing — energy consumption patterns, SEC benchmarking and energy-efficiency roadmap.